

Program: Diploma in Automation and Robotics	
Course Code: 5339C	Course Title: IoT and Applications Lab
Semester: 5	Credits: 3
Course Category: Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To introduce the raspberry PI platform, that is widely used in IoT applications
- To introduce various cloud platforms.

General Instructions:

- Students have to write the date, aim, Software & Hardware requirements for that experiment in the observation book.
- Students have to listen and understand the experiment explained by the faculty and note down the important points in the observation book.
- Students need to write procedure/algorithm in the observation book.
- Analyze and Develop/implement the logic of the program by the student in respective platform
- After approval of logic of the experiment by the faculty then the experiment has to be executed on the system.
- After successful execution the results are to be shown to the faculty and noted the same in the observation book.
- Students need to attend the Viva-Voce on that experiment and write the same in the observation book.
- Update the completed experiment in the record and submit to the concerned faculty in-charge.

Course Prerequisites:

Topic	Course code	Coursename	Semester
Basics of Arduino, Microcontroller	2371	Foundations of Robotics	2

Tools and Equipment:

Processors: Intel Atom® processor or Intel® Core™ i3 processor.

Disk space: 1 GB.

Operating systems: Windows 7 or later, macOS, and Linux.

Software: PyCharm, Arduino IDE

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	To use microcontroller based embedded platforms in IOT	12	Applying
CO2	To introduce the raspberry PI platform, that is widely used in IoT applications	12	Applying
CO3	To use wireless peripherals for exchange of data	9	Applying
CO4	To make use of Cloud platform to upload and analyze any sensor data	9	Applying
	Lab Exam	3	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			
CO5	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To use microcontroller based embedded platforms in IOT		

M1.01	Calculate temperature using a temperature sensor.	3	Applying
M1.02	Interfacing Arduino to Zigbee module	3	Applying
M1.03	Interfacing Arduino to GSM module	3	Applying
M1.04	Interfacing Arduino to Bluetooth Module	3	Applying
CO2	To introduce the raspberry PI platform, that is widely used in IoT applications		
M2.01	Familiarize with raspberry PI board and Raspbian OS	3	Applying
M2.02	Introduction to Raspberry PI platform and python programming	3	Applying
M2.03	Interfacing sensors to Raspberry PI	6	Applying
	Lab ExamI	1.5	
CO3	To use wireless peripherals for exchange of data		
M3.01	Communicate between Arduino and Raspberry PI using Bluetooth	3	Applying
M3.02	Communicate between Arduino and Raspberry PI using wifi	3	Applying
M3.03	Communicate between Arduino and Raspberry PI using Zigbee	3	Applying
CO4	To make use of Cloud platform to upload and analyze any sensor data		
M4.01	Setup a cloud platform to log the data.	3	Applying
M4.02	Log Data using Raspberry PI and upload to the cloud platform	6	Applying
	LabExam II	1.5	

Online Resources:

Sl.No	WebsiteLink/Textbooks
1	Internet of Things - A Hands-on Approach, ArshdeepBahga and Vijay Madiseti, Universities Press, 2015, ISBN: 9788173719547

2	Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD),2014, ISBN: 9789350239759.
3	Bernd Scholz-Reiter, Florian Michahelles, “Architecting the Internet of Things”, ISBN 978-3-642-19156-5 e-ISBN 978-3-642-19157-2, Springer, 2016
4	https://github.com/